DATA WAREHOUSE

A simple, exam-focused guide

Definition: A Data Warehouse (DW) is a centralized repository that collects, integrates, stores, and organizes data from multiple sources so that it can be used for reporting, analysis, business intelligence, and decision-making.

1. Why Do We Need a Data Warehouse?

• Organizations generate data from many systems such as sales applications, websites, ERP systems, CRM systems, databases, and spreadsheets.

• Operational databases are mainly designed for day-to-day transactions, while a data warehouse is designed for analysis of large amounts of historical data.

• A warehouse combines data from different sources into a consistent format, making it easier to compare performance and identify trends.

2. Data Warehouse Architecture

Typical flow: Data Sources → ETL/ELT → Staging Area → Data Warehouse → Data Marts / OLAP → Reports, Dashboards and Analytics

Layer	Purpose
Data Sources	Operational databases, files, APIs, ERP, CRM and other systems.
ETL / ELT	Extracts data, transforms/cleans it, and loads it into the warehouse.
Staging Area	Temporary area used to prepare and validate incoming data.
Data Warehouse	Central repository containing integrated and historical data.
Data Marts	Smaller subject-oriented stores for departments such as sales or finance.
OLAP / BI	Supports multidimensional analysis, dashboards and reports.

3. ETL Process

Extract: Collect data from source systems. **Transform:** Clean, standardize, validate, join, filter and convert data. **Load:** Store the prepared data in the warehouse.

Example: A company receives sales data from three branches using different date formats and currency representations. ETL converts them into a common format before loading the data into the warehouse.

4. Characteristics of a Data Warehouse

Characteristic	Meaning
Subject-oriented	Organized around subjects such as customers, products or sales.
Integrated	Combines data from different sources into a consistent format.
Time-variant	Maintains historical data so changes can be analyzed over time.

Non-volatile	Data is mainly read for analysis and is not frequently updated like OLTP data.

5. Data Warehouse vs OLTP

Feature	OLTP	Data Warehouse / OLAP
Main purpose	Daily transactions	Analysis and decision support
Data	Mostly current	Current + historical
Operations	Insert, update, delete	Mostly read/query
Queries	Short and frequent	Complex analytical queries
Design	Usually normalized	Often dimensional/denormalized
Users	Clerks, applications	Analysts, managers, data scientists

6. Dimensional Modeling

Dimensional modeling organizes warehouse data into **fact tables** and **dimension tables** for efficient analysis.

Fact Table: Contains measurable business values such as sales amount, quantity, profit and number of orders, along with keys linking to dimensions.

Dimension Table: Describes the context of a fact, such as customer, product, location and time.

7. Star Schema

A star schema has one central fact table connected directly to several dimension tables. It is simple and commonly used for analytical queries.

Dimension: Time	→	FACT: Sales	←	Dimension: Product
Dimension: Customer	→	Sales Amount / Quantity	←	Dimension: Store

8. Snowflake Schema

A snowflake schema is an extension of the star schema in which dimension tables are further normalized into related tables. It can reduce redundancy but may require more joins.

9. OLAP Operations

Operation	Meaning
Roll-up	Summarize data to a higher level, e.g., city → state → country.
Drill-down	Move from summarized data to more detailed data, e.g., year → quarter → month.
Slice	Select one dimension value, e.g., sales for 2026 only.
Dice	Select a smaller sub-cube using multiple dimension conditions.
Pivot	Rotate the view of dimensions to see data from another perspective.

10. Advantages of a Data Warehouse

• Provides a single, consistent view of organizational data.

• Supports historical trend analysis.

• Improves reporting and business intelligence.

• Separates analytical workloads from transaction-processing systems.

• Makes complex queries and dashboards easier to build.

• Helps management make data-driven decisions.

11. Limitations

• Initial implementation can be expensive and time-consuming.

• ETL/ELT pipelines require maintenance.

• Poor data quality in source systems can affect warehouse results.

• Large warehouses require significant storage and processing resources.

12. Simple Real-Life Example

Suppose an online retailer has separate systems for orders, customers, products and payments. A data warehouse collects data from all these systems. After ETL, an analyst can ask: *"What were the monthly sales of each product category in each region during the last three years?"* The warehouse can answer this using integrated historical data, while the operational systems continue handling normal customer orders.

13. Important Exam Points

• **Data Warehouse:** Central repository for integrated, historical data used mainly for analysis.

• **ETL:** Extract → Transform → Load.

• **Four key characteristics:** Subject-oriented, Integrated, Time-variant, Non-volatile.

• **Star schema:** One fact table directly connected to dimension tables.

• **Snowflake schema:** Normalized dimensions connected through multiple related tables.

• **OLAP:** Supports multidimensional analytical operations such as roll-up, drill-down, slice, dice and pivot.

One-line definition for exams: A Data Warehouse is a subject-oriented, integrated, time-variant and non-volatile repository of historical data designed to support analytical processing and decision-making.